Justification Document

Semester 6 Advanced Technology

C. Stijlaart

Table of Contents

1. Preface

In this justification document, I will provide a detailed explanation for each of my professional products developed during the group project and personal project of Semester 6 Advanced Technology. The document will showcase the systematic approach I took to address problems and opportunities, outline my goals and solutions, detail the questions I needed to answer, summarize my activities, describe how I validated my solutions, and conclude whether I successfully solved the problem.

2. Document history

Table 1. Document History

Version	Date	Status	Author	Description	Remarks
0.1	01-10-2023	Draft	Casey Stijlaart	Added Sprint 1 professional product	-
0.2	03-11-2023	Draft	Casey Stijlaart	Added Sprint 2 professional product	-
1.0	06-11-2023	Revised First Version	Casey Stijlaart	Revision based upon feedback C. van Tilborg	-
1.1	23-11-2023	First Version	Casey Stijlaart	Added Sprint 3 professional product	-
2.1	14-12-2023	Second Version	Casey Stijlaart	Added Sprint 4 professional product	Revision based upon feedback C. van Tilborg has been included in this update.
3.0	19-01-2024	Final Version	Casey Stijlaart	Added Sprint 5 professional product	-

3. General

3.1. Project Management - Git

3.1.1. File(s)

- PP+GP-project-management-git.pdf
- PP+GP-git-pipeline.pdf

3.1.2. Context and Problem/Opportunity

During the first sprint of our group project, our team faced challenges related to communication and leadership. There was a lack of clarity responsibilities and overall project direction. Therefore it became evident to me, that action was needed to address the issues.

3.1.3. Goal and Intended Solution

Recognizing the need for clear project management, I took the initiative to step and help the team guide in the initial start up phase. My goal was to establish a structured manner of work, that would guide the team and provide a clear project direction.

In parallel, understanding the importance of robust testing, documentation, and code quality in our project, I integrated a GitLab CI/CD pipeline.

3.1.4. Questions to Be Answered

To plan this solution, I needed answers to questions such as:

- How can better overview be given to the group, regarding to tasks?
- How can we ensure version control and traceability?
- How can we enhance code quality, testing, and documentation in our software development process?

3.1.5. Summary of Activities

With the help of the questions, a discussion was held to inform the group on the approach. The approach meant to make use of issues with labels, such as "next sprint backlog", "sprint backlog", "in progress", and "in review". Another approach was to make use of milestones, is it can give a clear insight on how far we are we certain aspects of out project. Version control was another urgent aspect. Together with M. Rats, I set up the branching and templates for issues and merge requests, and merge request rules. Second last, to help kick it off, I had taken it upon myself to create the issues we should be working on for the next sprint and helped with dividing them, ensuring the given issues corresponded with the research they had done for the project. Lastly, the CI/CD pipeline was integrated, to enhance our software developemnt process by providing a systematic and automated approach to testing, documentation, and code quality.

3.1.6. Validation of Solution

Utilizing the mentioned labels provided a better insight into our current goals and what’s next. It ensured that when tasks were completed, there was always a clear path forward. Regarding version control, rules were implemented to protect the main branch, 'Development'. These rules were added to the git project configuration to ensure compliance. An example of how issues, labels, and milestones have been used can be found in the following document: [GP+PP-project-management.pdf].

Simultaneously, the GitLab CI/CD pipeline ensured comprehensive testing, documentation generation, and code style checks. The modular structure allows for flexibility and easy maintenance, and the pipeline is configured to meet the specific needs of our project. An example of how the GitLab CI/CD pipeline has been used can be found in the following documents: [GP+PP-project-management.pdf] and [GP+PP-git-pipeline.pdf].

3.1.7. Conclusion and Recommendations

The workflow significantly improved due to the changes implemented. Everyone had better guidance and a more structured way of working, which ultimately enhanced the team’s effectiveness. The integration of the GitLab CI/CD pipeline ensures continuous improvement in code quality and development efficiency. Regularly reviewing and updating both the project management processes and the CI/CD pipeline is recommended to adapt to evolving project requirements and industry best practices.

3.2. Project Plan

3.2.1. File(s)

- PP-project-plan.pdf
- GP-project-plan.pdf

3.2.2. Context and Problem/Opportunity

A well-structured project plan is essential to provide a clear indication on the project and serves as guidance.

3.2.3. Goal and Intended Solution

The primary goal of this project plan is to create a roadmap for the personal project. By providing a structured approach, it intends to guide the project's phases, set clear objectives. The intended solution is to develop a comprehensive project plan that discusses all aspects of the project, from problem analysis to risk management.

3.2.4. Questions to Be Answered

To formulate this project plan, several questions needed to be addressed:

- How can a project plan effectively address the needs of a personal project, considering its characteristics and goals?
- What strategies and methodologies are best suited to guide the project from initiation to completion?
- What measures should be in place to identify and manage potential risks during the project's execution?

3.2.5. Summary of Activities

The creation of the project plan involves various activities, including the analysis of the project's context, the establishment of a structured approach, and the formulation of a timeline and team organization. It also required the identification and assessment of risks and the fallback measures.

3.2.6. Validation of Solution

Validation of this solution was achieved through practical application. The project plan was created, serving as the guiding document for the personal project. The complete documents [PP-project-plan.pdf] and [GP-project-plan.pdf] can be referred to for detailed insights and guidance.

3.2.7. Conclusion and Advises

In conclusion, this project plan serves as a critical tool for personal project management, bridging the gap between project concept and completion. It provides a structured approach, ensuring that the project's goals are met.

3.3. Test Applications

3.3.1. File(s)

- PP-Test-Application.png
- GP-Test-Application.png
- GP-test-application-automated-connection.mp4
- GP-Application-Database-Access-Control.mp4
- GP-Dashboard-Experiment-Control.mp4
- PP-Musical-Composition-Control.mp4

The development of a test application serves an important role for both the personal project as the group project. A test application offers a practical means of acquiring proficiency with the database, MongoDB, and validating its suitability for project requirements.

3.3.2. Significance of Test Application

The significance of this test application lies in its ability to mimic real-world scenarios, allowing project stakeholders to gain hands-on experience with the database. It allows for the team to assess the database’s capabilities and determine whether it aligns with the project’s needs, all while offering a controlled environment for experimentation.

3.3.3. Differentiated Functionality

While the test application shares a common objective in promoting MongoDB understanding, it features differentiated functionalities to cater to the distinct requirements of personal and group projects.

- **Group Project Test Application:** Primarily focuses on data retrieval based on group-specific views, aligning with the group project’s objectives.
- **Personal Project Test Application:** Encompasses additional features such as data insertion, retrieval, and collection management, addressing the requirements of personal projects.

3.3.4. Rationale

The rationale behind creating these test applications is rooted in the need for a practical and effective approach to familiarize project participants with MongoDB’s operation and assess its suitability. It bridges the gap between theory and practice, offering a platform for hands-on learning and validation.

3.3.5. Questions Addressed

The development of the test application aimed to address several critical questions:

- How can a test application realistically simulate the database usage, providing an clear indication of its way of working for the project stakeholders?

- What specific functionalities should be integrated into the test application to showcase the full spectrum of requirements for both personal and group projects?
- How can the test application allow for a comprehensive understanding of the databases operations, including data retrieval, insertion, and collection management?

3.3.6. Practical Implementation

Creating the test application involved a series of practical activities, including the analysis of project requirements, the design and development of the application with tailored features, and testing to ensure it accurately replicates real-world database usage.

3.3.7. Validation and Future Considerations

The validation of this solution was achieved through practical application and testing. Real-world interactions with the database, facilitated by the test application, confirmed its effectiveness in familiarizing users and verifying MongoDB’s alignment with project requirements.

3.4. System Design Document

3.4.1. File(s)

- PP-system-design-document.pdf
- GP-system-design-document.pdf

3.4.2. Context

This justification document is created to explain the purpose and significance of the System Design Documents for the personal- and group project. The documents details the design and architecture of an embedded system.

3.4.3. Problem/Opportunity

The complexity of the embedded system, involving multiple components and intricate interactions, demands a systematic and well-documented design approach. These documents outlines the reasons behind creating the System Design Document and its role in ensuring a structured and effective development process.

3.4.4. Goal and Intended Solution

The goal of the System Design Document is to provide a comprehensive guide that details the design and architecture of the embedded system. It aims to articulate the functionalities, interactions, and hardware components of the system, ensuring a clear roadmap for the development team.

The intended solution is a well-structured document that includes system descriptions, design components, and behavioral specifications. By incorporating diagrams such as hardware connections, class diagrams, and sequence diagrams, the document ensures a visual representation of the system's architecture. This aids in effective communication, development, and future system understanding.

3.4.5. Questions to Be Answered

- What components are there, and what is their purpose?
- In what manner will the various components be connected?
- Which type of design patterns are best suitable for the class diagrams?

3.4.6. Summary of Activties

The creation of this system design documents involved a series of activities, including the use cases, requirements, sequence diagrams and class diagram. Each of these activties playing a pivotal roll in a complete design.

3.4.7. Validation of Solution

Validation of this solution was achieved through practical application. The project plan was created, serving as the guiding document for the personal project. The complete documents, [PP-system-design-document.pdf] and [GP-system-design-document.pdf] can be referred to for detailed insights and guidance.

3.4.8. Conclusion and Recommendations

In conclusion, the System Design Document is a critical asset for the successful development of the embedded system. Regular updates to the document, especially after key development milestones, are of importance to ensure its alignment with the evolving project requirements. Additionally, collaboration and feedback from the development team should be encouraged to enhance the document's accuracy and effectiveness as a guide for system implementation.

3.5. Project Report

3.5.1. File(s)

- GP-project-report.pdf
- PP-project-report.pdf

3.5.2. Context and Problem/Opportunity

At the end of a project, it is of great necessity to reflect on the journey, outcomes, and lessons learned during the development of an embedded software project. The creation of a comprehensive project report serves as a crucial component in documenting the project's evolution, findings, and areas for improvement.

3.5.3. Goal and Intended Solution

The primary goal of the project report is to provide a retrospective analysis of the completed project. It aims to offer insights into the background, objective, scope, and most importantly, the findings encountered throughout the project. The report serves as a platform to identify opportunities for enhancement and suggests viable solutions for overcoming limitations.

3.5.4. Questions to Be Answered

- What were the key findings and challenges encountered during the project's lifecycle?
- How can the project be improved to enhance its overall performance, efficiency, and effectiveness?

3.5.5. Validation of Solution

The validation of the solution lies in the comprehensive documentation within the project report. This includes a detailed examination of the identified findings, proposed solutions, and any validations or testing performed to ensure the effectiveness of the suggested improvements.

3.5.6. Conclusion and Recommendations

The project report concludes by summarizing the key takeaways from the project, emphasizing the significance of the findings, and providing actionable recommendations for future endeavors. These recommendations may include best practices, potential enhancements, or considerations for addressing challenges that arose during the project. Overall, the project report serves as a valuable resource for stakeholders, enabling them to learn from the project's experiences and apply these insights to future embedded software projects.

4. Group Project

4.1. Database and Data Structure Research

4.1.1. File(s)

- GP-research-database.pdf

4.1.2. Context and Problem/Opportunity

Database storage and data structure selection are an important decision in any software project. Different options have their own set of advantages and disadvantages, and it's crucial to choose the most suitable ones for our project.

4.1.3. Goal and Intended Solution

The goal of this research was to examine multiple database storage options and possible data structure techniques before making any decisions. Decisions that align with our project's specific requirements.

4.1.4. Questions to Be Answered

To plan this solution, different questions had to be answered, such as : - What are the pros and cons of local and cloud-based databases? - Which data structure types are most suitable for our use case? - How do these choices impact project scalability, performance, and maintenance?

4.1.5. Summary of Activities

I conducted thorough research on database storage options, focusing on local and cloud-based databases. This research led to a recommendation that considers our project's goals and stage. Additionally, I explored data structure techniques, including tables, normalization, and indexes, and their applications in the context of our database.

4.1.6. Validation of Solution

Validation was achieved by considering the practicality of the recommendations in the context of our project. For database storage, the choice was influenced by our project's nature as a prototype, and for data structures, the recommendations were based on scalability, performance, and data organization needs.

4.1.7. Conclusion and Advises

After careful consideration, I recommend using MongoDB, a NoSQL database, for our prototype. MongoDB offers advantages in data control, performance, and ease of setup. Additionally, it provides robust security options that can be implemented to safeguard our data. For data organization, I recommend employing tables with normalization techniques combined with indexes to enhance performance. These choices align with our project's immediate needs and future scalability.

4.2. Code Examples

4.2.1. File(s)

- GP-Code-Examples.pdf

4.2.2. Rationale

The "Code Examples" presents a comprehensive Docker container and database implementation for an IoT measurement setup designed to observe lab setups. The rationale behind this professional product is to provide a standardized and easily deployable solution for managing databases, complete with essential scripts and classes for effective data handling.

The Docker container encapsulates the project environment, ensuring reproducibility and ease of deployment. It incorporates versioning (V1.5) and includes MongoDB drivers, a running MongoDB server, Google tests, code coverage through lcov/gcov, and more. This version serves as a solid foundation for future enhancements and additions.

The database implementation is structured around three distinct classes - "Database," "DatabaseDataFormatter," and "DatabaseManager.". This design choice adheres to SOLID principles, particularly focusing on loose coupling and single responsibility. The database classes facilitate various operations, from interacting with the database server to formatting data and managing the overall application-to-database interaction.

4.2.3. Questions Addressed

The project addresses several key questions in its design and implementation:

- How is the Docker container structured?

The Docker container consists of a Dockerfile and two scripts, "install.sh" and "start-script.sh." The former handles package installation and updates, while the latter serves as the entrypoint script, setting up MongoDB with appropriate users and security.

- What are the key components of the database implementation?

The database is comprised of three classes: "Database," "DatabaseDataFormatter," and "DatabaseManager." Each class has a specific role, promoting separation of concerns and maintaining a modular and extensible codebase.

- How is testing integrated into the project?

The document includes a demonstration of unit testing with the DatabaseManager as an example. Testing ensures the reliability and correctness of the implemented functionality, with a focus on dependency injection to mock the database for effective unit testing.

4.2.4. Practical Implementation

Docker Container

The Docker container simplifies the deployment and management of the project environment. Users can easily set up the required packages and drivers using the provided "install.sh" script. The "start-script.sh" script initializes the MongoDB server with necessary configurations, making the container ready for use.

Database Classes

The three database classes (Database, DatabaseDataFormatter, DatabaseManager) are implemented following best practices such as Dependency Injection and the Strategy pattern. This allows for flexibility in handling various commands, maintaining loose coupling, and ensuring the separation of concerns.

Testing

Unit testing is an integral part of the project, as demonstrated by the DatabaseManagerTest. The test cases showcase the robustness of the DatabaseManager’s functionality, and the use of mock databases ensures thorough testing without affecting the actual database.

4.2.5. Validation and Future Considerations

The project has undergone validation through unit testing, ensuring that key components such as the Docker container and database classes function as intended. Future considerations for the project involve continuous improvement, including updates to the Docker image as additional packages are required. The modular structure of the database implementation also allows for easy extension and modification to accommodate new features or changes in requirements.

In conclusion, the "Code Examples: Groups Project" by Casey Stijlaart stands as a well-structured and thoughtfully designed professional product. Its Docker container and database implementation provide a solid foundation for managing IoT measurement setups, emphasizing modularity, reliability, and ease of deployment.

4.3. Message Protocol

4.3.1. File(s)

- GP-Message-Protocol.pdf

4.3.2. Context

The Messaging Protocol Document is a critical component within the larger framework of an IoT Measurement Setup. In this context, effective communication and coordination among Sensor Units, Control Units, and Camera Units are pivotal for the success of the overall system.

4.3.3. Problem/Opportunity

As there are various messages to be exchanged between the components, it is of great importance to map out what these messages are. Creating such a document also plays a pivotal role in ensuring every component sends the correct message for the other components, ensuring no miscommunication can occur.

4.3.4. Goal and Intended Solution

The overarching goal is to design a messaging protocol that enables seamless communication, ensuring that Sensor Units, Control Units, and Camera Units can exchange commands, data, and status updates in a standardized and efficient manner. The solution for this is a well-defined messaging protocol outlined in this document. The protocol utilizes a JSON-based message format, providing a clear structure for communication. Predefined messages cater to a range of functionalities, offering a versatile solution to the communication needs within the IoT Measurement Setup.

4.3.5. Questions to Be Answered

- How does the messaging protocol handle different units and their specific commands?

The document answers this question by providing examples of messages for various scenarios, such as retrieving sensor data, starting and stopping experiments, and reporting events.

- What is the message format, and how does it ensure clarity and flexibility?

The document addresses this question by presenting the JSON-based message format, breaking down each component to illustrate how it facilitates effective communication.

- How practical is the implementation of the messaging protocol in real-world scenarios?

The document answers this question by providing practical examples of request-response pairs and event reporting, demonstrating the protocol’s usability in diverse situations.

4.3.6. Summary of Activities

The document outlines the messaging protocol’s structure, format, and predefined messages. It provides practical examples to guide developers and stakeholders in implementing and integrating the protocol into their systems. Additionally, the document categorizes messages, making it a comprehensive guide for anyone involved in the IoT Measurement Setup.

4.3.7. Validation of Solution

The validation of the messaging protocol involves testing its functionality in simulated and real-world environments. This includes ensuring that messages are correctly interpreted and responded to by the respective units. Continuous validation efforts may involve user feedback and potential updates to accommodate evolving IoT technologies.

4.3.8. Conclusion and Recommendations

In conclusion, the Messaging Protocol Document is a crucial reference for achieving effective communication within an IoT Measurement Setup. The document provides a clear understanding of the protocol's design, implementation, and practical application. Future recommendations involve periodic reviews and updates to the messaging protocol to accommodate emerging requirements and technologies within the IoT landscape.

4.4. Project Configuration

4.4.1. File(s)

- GP-project-configuration.pdf

4.4.2. Context

The project involves the configuration of a group project aimed at providing a clear organization and promoting loose coupling between system components. The configuration includes an elaborate folder structure and utilizes CMake for streamlined project configuration, building, testing, and packaging.

4.4.3. Problem/Opportunity

The need for a well-defined project configuration arises to ensure organizational clarity, minimize interdependencies, and facilitate efficient project development and maintenance.

4.4.4. Goal and Intended Solution

The goal is to implement a project configuration that fosters loose coupling and organizational clarity. The intended solution includes an organized system folder structure and the use of CMake for project configuration, building, testing, and packaging.

4.4.5. Questions to Be Answered

1. How does the folder structure contribute to loose coupling between system components?
2. What role does CMake play in project configuration, building, testing, and packaging?
3. How are build flags implemented, and how do they facilitate selective target compilation?

4.4.6. Summary of Activities

- Define a system folder structure with "libs" and "targets" components.
- Utilize CMake for project configuration, building, testing, and packaging.
- Implement build flags to facilitate selective target compilation.

4.4.7. Validation of Solution

The solution will be validated by assessing the clarity of the system folder structure, the effectiveness of CMake in project configuration, and the successful use of build flags for selective target compilation.

4.4.8. Conclusion and Recommendations

In conclusion, the project configuration, consisting of a well-defined folder structure and the use of CMake, contributes to organizational clarity and efficient project development. The implementation of build flags further enhances configurability of the project.

5. Personal Project

5.1. Algorithm Research

5.1.1. File(s)

- PP-research-algorithms.pdf

5.1.2. Context and Problem/Opportunity

The project's central focus is to utilize algorithms for creating musical output, making algorithm research a critical starting point. The goal was to explore various algorithmic possibilities to identify the most suitable ones that would effectively demonstrate the project's key features.

5.1.3. Goal and Intended Solution

My primary aim was to push the boundaries of what I know. As my knowledge of algorithms within embedded software is rather small, I got pushed out of my comfort zone rather quickly. Many options came across my path, some looking easier and some looking more complexer then the others. Therefor, the balance in being able to bring my project idea to life and pushing my limits, had to be found. Besides, I intended to select algorithms that would best highlight the differences in their functionality.

5.1.4. Questions to Be Answered

To plan my path to the solution, I needed to answer questions such as: - What algorithms are relevant to my project? - How do these algorithms differ in functionality? - Which algorithms align with my project's purpose?

5.1.5. Summary of Activities

I conducted an extensive literature review and gathered information from the community on various algorithms. This research resulted in the creation of a research section within the research document, the section can be found in the [sprint-1/personal-project/research-algorithms.pdf].

5.1.6. Conclusion and Advises

In conclusion, this research offers a great understanding of various algorithmic techniques for generating music based on environmental data. To showcase diversity in output, I concluded in combining algorithmic composition, sonification, and data-driven soundscapes in the project.

5.2. Test Plan

5.2.1. File(s)

- PP-test-plan.pdf

5.2.2. Context

This justification document is created to provide a rationale for the Test Plan developed for the Personal Project. The project involves the design of an embedded system that transforms environmental data into musical compositions.

5.2.3. Problem/Opportunity

The complexity and interconnected nature of the system components necessitate a comprehensive testing approach to ensure the functionality, reliability, and performance of the embedded system. This document outlines the reasons behind the creation of the Test Plan and its importance in the project's success.

5.2.4. Goal and Intended Solution

The goal of the Test Plan is to systematically validate the embedded system's key functionalities, ensuring accurate environmental data collection, correct musical composition generation, seamless algorithm switching, and reliable database interactions.

The proposed solution is a structured testing approach that encompasses various testing types, such as unit testing, component testing, and black box testing. The use of Google tests for unit testing, component testing to verify individual components, and functional testing for black box testing ensures a thorough examination of the system from different perspectives.

5.2.5. Questions to Be Answered

To create a test plan, various aspects must be considered. As each project has its own purpose and different purpose, a test plan as made for just that specific project. While creating the test plan for this project, different factors must be addressed. To adress these, the following question needs to be answered.

- Which scenario's should be tested?
- What types of testing will be used?
- What units should be tested?

5.2.6. Summary of Activities

The creation of this test plan involved a series of activities, including the creation of various test scenario's. Besides that, it includes what kind of testing methods are used and for what it will be used.

5.2.7. Validation of Solution

The chosen testing types collectively provide a robust validation approach. Unit tests ensure the correctness of individual components, component tests validate component interactions, and black box testing ensures alignment with specified requirements. The combination of these testing methodologies aims to identify and address potential issues throughout the development lifecycle.

5.2.8. Conclusion and Recommendations

In conclusion, the Test Plan is a crucial component of the project's development process, ensuring the reliability and functionality of the embedded system. To enhance the effectiveness of the testing process, it is recommended to regularly update the plan based on project progress and continuously refine test cases to adapt to evolving system requirements. Additionally, the integration of testing into the development pipeline ensures that any issues are promptly identified and resolved, contributing to a more stable and robust final product.

5.3. Generation of Musical Composition

5.3.1. File(s)

- PP-Output-CustomComposition-Poor.mp3
- PP-Output-CustomComposition-Good.mp3

5.3.2. Rationale

This professional product is designed to demonstrate the transformative power environmental data, in creating musical compositions. The rationale behind this project is to explore the intersection of data science and creative arts, providing an innovative way to represent and interpret environmental conditions through music.

The project leverages the concept that environmental data quality, such as humidity levels, influences the mood and characteristics of the musical composition. This synthesis of scientific data and artistic expression aims to create an immersive experience for users, allowing them to perceive and appreciate environmental conditions through the auditory medium.

5.3.3. Questions Addressed

- How does environmental data influence the musical composition?

The project addresses this question by showcasing two distinct musical compositions. The first composition reflects a period of poor environmental data, resulting in a more intense and darker musical output. Conversely, the second composition corresponds to a period of overall good environmental data, resulting in a brighter and calming musical output.

5.3.4. Practical Implementation

Musical Composition 1: Poor Environmental Data (PP-Output-AlogorithmicComposition-Poor.mp3)

The first file showcases a musical composition influenced by overall poor environmental data, with occasional moments of better data quality. The composition reflects the project's commitment to accurately translating data nuances into musical elements. For instance, the use of heavy and dark tones during poor environmental conditions emphasizes the immersive connection between data and artistic output.

Musical Composition 2: Good Environmental Data (PP-Output-AlogorithmicComposition-Good.mp3)

The second file presents a contrasting musical composition influenced by consistently good environmental data. The composition aims to evoke a sense of brightness and calmness, aligning with the positive environmental conditions. This serves as a testament to the flexibility of the project in adapting the musical output based on varying data inputs.

5.3.5. Validation and Future Considerations

In conclusion, the product successfully connects data science and artistic expression, offering a novel way to experience and interpret environmental conditions through music. Future considerations for the project involve expanding the range of environmental parameters considered and refining the mapping of data to musical elements.